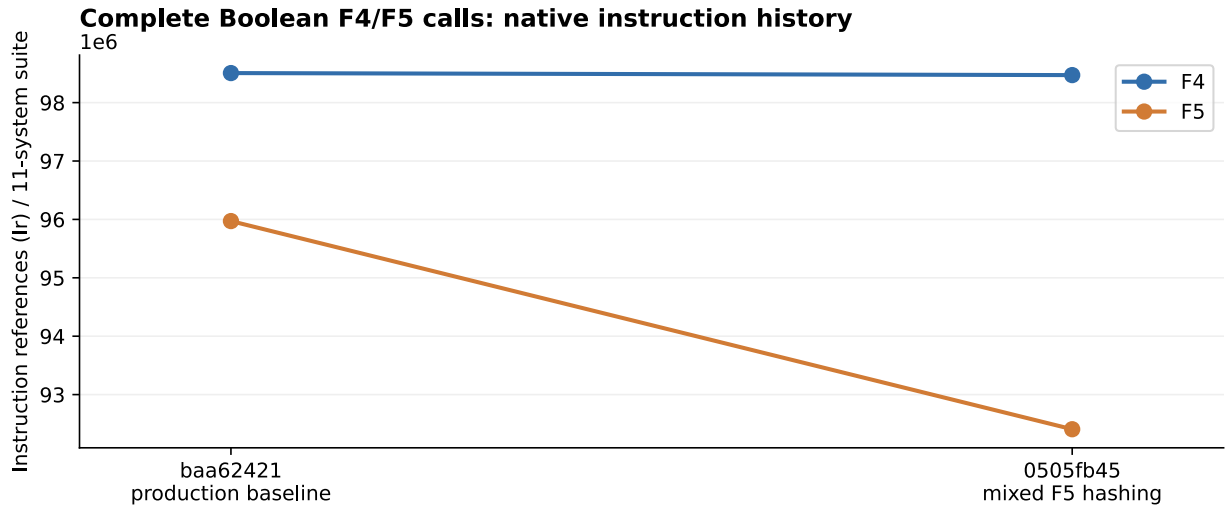
Complete Boolean F4/F5 calls

Native instruction and allocation measurements · 2026-10-12



5 paired A/B/A′ rounds; native medians and all samples. Merged worker coverage.
Instrumented complete calls; no wall-time or RSS claim. Source: records/comparison.json.

F4: 98,506,024 → 98,471,390 Ir per suite. Paired instruction reduction 0.04% (95% bootstrap interval 0.029% to 0.042%). Native verdict: neutral.

F5: 95,971,529 → 92,406,706 Ir per suite. Paired instruction reduction 3.71% (95% bootstrap interval 3.714% to 3.726%). Native verdict: fewer instructions.

Five paired rounds of all eleven frozen systems, one Rayon worker. F4 is a control. The native verdict requires a 95% bootstrap interval excluding zero and an effect above twice the A/A noise estimate and 2%. Every reduced output, input hash, pruning report and word-XOR count matched. No per-workload instruction regression crossed the gate.

Native sources: comparison and raw-record hashes, frozen protocol, result report and validation. Raw Callgrind dumps and Rust JSONL remain beside the source data. This 11-system suite is separate from historical n=17/n=23 CI diagnostics. Validation: 3,906 release-library tests passed, 110 ignored, zero failures. The previously unfinished test passed unmodified. All 18 focused F5 tests and five comparison tests passed. Runtime and process RSS remain unmeasured.